

AERO50003

Introduction to Numerical Methods for ODEs and PDEs

Tutorial Sheet 1

Finite Differences

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Exercise 1 (The first derivative of a sinusoidal function)

Consider the function

$$f(x) = \sin(x),$$

and the interval $x \in [0, 15]$. We wish to compute the derivative of $f(x)$ numerically. To do so, we create a grid of uniformly spaced points (for the variable x) from $x = 0$ to $x = 15$. Let this uniform spacing be some constant $h > 0$. We can label each point on the grid as $x_1, x_2, \dots, x_N$ (where, differently from the lecture notes, we start with the index 1 instead of the index 0 to be consistent with the syntax of Matlab).

1. Determine the derivative of $f(x)$ with respect to x analytically.
 - (a) Plot the derivative of the function for x in the interval of interest.
 - (b) Pick $h = 0.5$ (we will vary the grid sizing later) and plot (on the same figure), using asterisks, the (exact) derivative of $f(x)$ at each grid point.
2. We now wish to compute the derivative numerically using first order forward differencing with $h = 0.5$.
 - (a) Use first order forward differencing to approximate the derivative of the function at the gridpoints $x_1, \dots, x_{N-1}$.
 - (b) Plot, using an asterisk, the numerical approximation of the derivative computed in part (a).
3. We now wish to compute the derivative numerically using first order backwards differencing with $h = 0.5$.
 - (a) Use first order backwards differencing to approximate the derivative of the function at the gridpoints $x_2, \dots, x_N$.
 - (b) Plot, using an asterisk, the numerical approximation of the derivative computed in part (a).
4. We now wish to compute the derivative numerically using second order central differencing with $h = 0.5$.
 - (a) Use first order forwards differencing to approximate the derivative of the function at the gridpoint x_1 .
 - (b) Use second order central differencing to approximate the derivative of the function at the gridpoints $x_2, \dots, x_{N-1}$.
 - (c) Use first order backwards differencing to approximate the derivative of the function at the gridpoint x_N .
 - (d) Plot, using an asterisk, the numerical approximation of the derivative computed in parts (a)-(c).

5. Repeat parts 1-3 with different grid spacing h and compare the “numerical approximations” with the derivative calculated analytically for the different values of h . In particular, let the gridspacing vary from 0.1 to 1 (feel free to experiment with values outside of this range too!). What do you observe? (It may be beneficial to plot the derivative and the various numerical approximations on the same figure.)

Exercise 2 (The second derivative of a sinusoidal function)

Consider the function

$$f(x) = \sin(x),$$

and the interval $x \in [0, 15]$. We wish to compute the second derivative of $f(x)$ numerically. Again, we create a grid of uniformly spaced points (with some uniform spacing $h > 0$) from $x = 0$ to $x = 15$. We can again label each point on the grid as $x_1, x_2, \dots, x_N$ (again, starting with index 1 is an arbitrary choice).

1. Determine the second derivative of $f(x)$ with respect to x analytically.
 - (a) Plot the second derivative of the function for x in the interval of interest.
 - (b) Pick $h = 0.5$ (we will vary the grid sizing later) and plot (on the same figure), using asterisks, the (exact) second derivative of $f(x)$ at each grid point.
2. We now wish to compute the second derivative of the function using second order central differencing with $h = 0.5$.
 - (a) Use second order central differencing to approximate the second derivative of the function at the grid points $x_2, x_3, \dots, x_{N-1}$.
 - (b) Plot the numerical approximation of the derivative computed in part (a).
 - (c) Plot (on the same figure), using asterisks, the (exact) second derivative of $f(x)$ at each grid point.
3.
 - (a) Repeat parts 1-2 with different grid spacing h and compare the “numerical approximations” with the derivative calculated analytically for the different values of h . What do you observe?
 - (b) In a new figure, plot the error at each of the gridpoint $x_2, \dots, x_{N-1}$ for three different stepsizes h (e.g. $h = 0.1$, $h = 0.5$ and $h = 1$).